

MAY 4 2026 SYLLABUS — 7 DOMAINS — ZERO DUPLICATE EDITION

DATABRICKS CERTIFIED DATA ENGINEER ASSOCIATE

144 Verified Unique Questions | 100 Foundational + 44 New 2026 Topics | Syllabus + Key Topics Cheat Sheet

144

Unique Questions

7

Exam Domains

45

Real Exam Qs

70%

Pass Mark ~32/45

QUESTIONS

45 scored MCQs

TIME LIMIT

90 minutes

PASS MARK

~70% (32/45 correct)

FEE

USD 200 + taxes

VALIDITY

2 years

DELIVERY

Online proctored / test centre

#	Domain	%	~Qs (45)	Min Correct	2026 Changes
1	Databricks Intelligence Platform	10%	5	4	Delta Lake, Liquid Clustering, Predictive Opt. updated
2	Data Ingestion & Loading	20%	9	6	Lakeflow Connect added, COPY INTO focus
3	Data Transformation & Modeling	18%	8	6	LDP/SDP 2026 API (@dp.* decorators)
4	Working with Lakeflow Jobs	12%	5	4	Data-driven triggers, streaming modes
5	Implementing CI/CD	10%	5	4	NEW DOMAIN: DABs, CLI, Repos, Databricks Connect
6	Troubleshooting, Monitoring & Optimization	15%	7	5	NEW: Spark UI, Liquid Clustering, AQE, disk spill
7	Governance & Security	15%	6	4	EXPANDED: ABAC, Column Masking, DENY, Row Filters
TOTAL		100%	45	~32	

Week 1: Domain 1 — Delta Lake: ACID, time travel, OPTIMIZE, VACUUM, Liquid Clustering, Predictive Optimization, compute types

Week 2: Domain 2 — Auto Loader, COPY INTO, Lakeflow Connect, JDBC, schema enforcement/evolution, medallion architecture

Week 3: Domain 3 — PySpark transforms, SQL functions, DLT/LDP expectations, SCD Type II, 2026 dp.* API

Week 4: Domain 4 — Lakeflow Jobs: triggers, streaming modes, watermark, checkpoint, output modes, retries

Week 5: Domains 5+6 — DABs, CI/CD, Spark UI debugging, Liquid Clustering, AQE, OOM, disk spill

NEW

Week 6: Domain 7 + Full mock — UC governance, ABAC, column masking, row filters, Delta Sharing, Lakehouse Federation

EXPANDED

PAGE 2 — KEY TOPICS CHEAT SHEET (All 7 Domains)

DOMAIN 1: Databricks Intelligence Platform

Topic	Key Point
Lakehouse Architecture	Open formats (Parquet/Delta) on cloud storage + ACID transactions + schema enforcement + BI/ML on same data. Eliminates separate data lake + warehouse layers.
Delta Lake / _delta_log	Data stored as Parquet. Transaction log in _delta_log/ (JSON per commit). Provides ACID, schema enforcement, time travel, CDF.
All-Purpose vs Job Cluster	All-purpose: persistent, interactive, expensive. Job cluster: created per run, auto-terminates — cost-efficient for production jobs.
SQL Warehouse / Serverless	SQL Warehouse: purpose-built for SQL/BI, auto-scales. Serverless: instant start, zero config, billed per DBU consumed.
Photon Engine	Databricks C++ vectorized engine. Accelerates SQL/DataFrame (aggregations, joins) without code changes.
Time Travel	SELECT * FROM t VERSION AS OF 5 or TIMESTAMP AS OF '2026-01-01'. RESTORE TABLE t TO VERSION AS OF 5 to revert.
OPTIMIZE / VACUUM	OPTIMIZE: compacts small files. VACUUM: removes unreferenced files older than retention (default 7 days). Always OPTIMIZE first, then VACUUM.
Liquid Clustering (NEW)	CLUSTER BY (col) replaces OPTIMIZE+ZORDER. Incremental reclustering, handles high-cardinality, flexible key changes. No full table rewrite needed.
Predictive Optimization (NEW)	Auto-runs OPTIMIZE+VACUUM on UC managed tables based on usage. No manual maintenance scripts needed.
Change Data Feed (CDF)	ALTER TABLE t SET TBLPROPERTIES ('delta.enableChangeDataFeed'='true'). Read changes: table_changes(). Row types: insert/update_preimage/update_postimage/delete.

DOMAIN 2: Data Ingestion & Loading

Topic	Key Point
Auto Loader (cloudFiles)	.format("cloudFiles"). Incremental with checkpoint. Directory listing (default) or file notification mode (SQS/Event Grid, scales to billions of files).
Key Auto Loader Options	schemaLocation (persist schema), inferColumnTypes=false (all strings), rescuedDataColumn (capture mismatches), schemaEvolutionMode=addNewColumns, maxFilesPerTrigger.
COPY INTO	Idempotent SQL command. Tracks loaded files in Delta log — safe to re-run. Best for moderate volumes/one-time loads. Not streaming.
Auto Loader vs COPY INTO	Auto Loader: Structured Streaming, billions of files, event-driven. COPY INTO: SQL, simpler, idempotent, moderate volumes.
mergeSchema vs overwriteSchema	mergeSchema=true: adds new columns, keeps existing. overwriteSchema=true: replaces entire schema with DataFrame's schema.
JDBC Ingestion	spark.read.jdbc(). Parallel: partitionColumn + lowerBound + upperBound + numPartitions creates N sub-queries in parallel.
Lakeflow Connect (NEW)	Managed connectors for Salesforce, SAP, databases. Handles auth, incremental sync, schema mapping, error recovery, UC governance automatically.

DOMAIN 3: Data Transformation & Modeling

Topic	Key Point
Medallion Architecture	Bronze=raw as-is (append-only). Silver=cleansed/validated/typed. Gold=aggregated KPIs/ML features. Gold feeds BI dashboards and ML models.
2026 SDP API	from pyspark import pipelines as dp. @dp.table (streaming, readStream). @dp.materialized_view (batch, dp.read()). @dp.temporary_view (staging, not persisted).
DLT/LDP Expectations	@dlt.expect=warn+keep row. @dlt.expect_or_drop=drop row, continue. @dlt.expect_or_fail=halt pipeline. Same as @dp.* in 2026 SDP.
SCD Type II (2026)	dp.create_streaming_table() + dp.create_auto_cdc_flow(stored_as_scd_type=2). Active records: __END_AT IS NULL. Old records: __END_AT = change timestamp.
Key SQL Functions	COALESCE(first non-null). NULLIF(null if equal). YEAR/EXTRACT/DATE_PART (all valid). PIVOT(rows→cols). UNPIVOT(cols→rows). NVL2(val, if_not_null, if_null).
Key PySpark Methods	withColumn(add/replace), withColumnRenamed, filter=where, union(stack all rows+dupes), dropDuplicates, repartition(full shuffle), coalesce(decrease only).
Streaming Limitations	ROW_NUMBER() NOT supported on streams → use dropDuplicates(['key']). monotonically_increasing_id() NOT supported on streams → use natural keys.
Broadcast Join	broadcast(small_df). Eliminates shuffle of large table. Auto-broadcast when < spark.sql.autoBroadcastJoinThreshold (default 10MB).

DOMAIN 4: Working with Lakeflow Jobs

Topic	Key Point
Trigger Types	Scheduled (cron). File Arrival (new file in path). Table Update (Delta table committed — best for data-driven chaining). Manual.
Streaming Triggers	ProcessingTime('30 seconds'): micro-batch every 30s. Once(): all data, 1 batch, stop. AvailableNow(): all data, multi-batch, stop. Continuous('100ms'): sub-second latency.

Output Modes	Append: new rows only (non-agg queries). Update: changed rows only (agg with watermark). Complete: full result every trigger (all aggs).
Watermark	withWatermark('ts', 'duration'). Bounds state for late data. Data within watermark threshold is included. Beyond it is dropped.
Checkpoint	Stores stream progress (offsets, state). Must be in cloud storage (S3/ADLS/GCS). Without it → restarts from beginning on failure.
Repair Run	Reruns only failed/skipped tasks. Preserves successful outputs. Most efficient recovery. max_retries=2 → 3 total attempts (1+2).
Task Dependencies	ALL_SUCCESS (default): task only runs if ALL upstream succeed. Failed upstream → downstream is skipped/failed. Conditional branching also supported.
dbutils.secrets	dbutils.secrets.get(scope, key). Never shown in notebook output. Backend: Azure Key Vault or Databricks-managed vault.

DOMAIN 5: Implementing CI/CD (NEW May 2026)

Topic	Key Point
Declarative Automation Bundles	Formerly Databricks Asset Bundles. Root: databricks.yml. YAML-based IaC for jobs, pipelines, notebooks. Deploys to dev/staging/prod with variable overrides.
DAB CLI Commands	validate (YAML check, no deploy). deploy (push resources). run <key> (trigger execution). destroy (remove workspace resources). Always validate before deploy.
DAB Targets	targets block in databricks.yml. Each target (dev/prod) overrides variables (catalog, cluster size). Same codebase everywhere — only config differs.
Databricks Repos	Git integration in workspace UI. Branch, commit, push, PR — works with GitHub, GitLab, Azure DevOps, Bitbucket.
CI/CD Workflow	commit → PR → review → merge to main → GitHub Actions: bundle validate → bundle deploy → (optional) bundle run.
Databricks Connect	SDK for local IDE (VS Code, PyCharm) that routes Spark execution to remote cluster. Local debugging + IntelliSense with full cluster power.

DOMAIN 6: Troubleshooting, Monitoring & Optimization (NEW May 2026)

Topic	Key Point
Data Skew	Few tasks much longer than others. Fix: AQE skew join, salting the skewed key, repartition by different column.
Disk Spill	Executor memory insufficient → data overflows to local disk (10-100x slower). Visible as 'Spill (disk)' in Stages tab. Fix: increase executor.memory, reduce partition size.
GC Time	High GC = JVM memory pressure. Fix: increase spark.executor.memory. Reduces frequency of garbage collection cycles.
Driver OOM	df.collect() on large data, broadcast table too large. Fix: avoid collect() on large data, increase spark.driver.memory.
Spark UI Key Tabs	Stages: task metrics, skew, spill, shuffle. SQL/DataFrame: physical plan DAG with exchange nodes. Executors: per-executor memory + GC. Streaming: batch latency.
AQE	spark.sql.adaptive.enabled=true (default Spark 3+). Runtime plan adjustments: coalesces partitions, fixes skew, switches join strategies automatically.
Liquid Clustering	CLUSTER BY col. Handles high-cardinality (no millions of directories). Incremental reclustering. Flexible key changes without full table rewrite.
shuffle.partitions Tuning	Default 200. Too many for small data = scheduling overhead > computation. Rule: ~128MB per partition. Reduce for small data, increase for large.

DOMAIN 7: Governance & Security (EXPANDED May 2026)

Topic	Key Point
UC Minimum Privileges	USE CATALOG + USE SCHEMA + SELECT to query a table. Missing any one blocks access. Inheritance: grant on catalog covers all schemas/tables below.
Managed vs External Tables	Managed: DROP TABLE deletes data. External (LOCATION keyword): data persists after DROP TABLE.
GRANT / REVOKE / DENY	GRANT privilege ON object TO principal. REVOKE removes grant (group membership may still allow access). DENY explicitly blocks — overrides GRANT even via group.
Column Masking (NEW)	SQL function at query time: IF(is_member('HR'), salary, NULL). Non-privileged users see masked values; HR managers see full values.
Row-Level Security (NEW)	Boolean function filtering rows per user/group using current_user() or IS_MEMBER(). Implement via dynamic views or UC ABAC policies.
ABAC Policies (NEW)	Centralised attribute-based policies combining row filters + column masks. Apply consistently across multiple tables based on user attributes.
Delta Sharing	Open protocol, zero-copy, live data. No Databricks needed by recipient. Read-only. Cross-cloud = egress fees. CREATE SHARE → ADD TABLE → CREATE RECIPIENT → GRANT.
Lakehouse Federation	Foreign catalogs → external DBs (MySQL, Snowflake). Query in place, no data movement. Predicate pushdown to source system.
Audit Logs	Workspace events → customer cloud storage (S3/ADLS/GCS) as JSON. Covers access, permissions, job runs. Used for GDPR/SOC2 compliance.
Data Lineage	UC auto-captures column-level lineage across all Databricks surfaces. No code changes. View in Catalog Explorer.

Q1-Q100: FOUNDATIONAL QUESTIONS (from practice exam paper)

Q1 FOUNDATIONAL

What is the primary advantage of a Lakehouse architecture over a traditional Data Warehouse?

Domain 1: Databricks Intelligence Platform

- A. It eliminates the need for data modeling
- B. It supports open file formats with ACID transactions and both BI and ML workloads ✓**
- C. It is always cheaper to operate
- D. It removes the need for a schema

Answer: B | **Why:** A Lakehouse combines low-cost open storage (like a Data Lake) with ACID transactions and governance from a Data Warehouse, supporting both BI and ML workloads on the same data.

Q2 FOUNDATIONAL

Which component of Databricks provides unified governance over data, analytics, and AI assets?

Domain 1: Databricks Intelligence Platform

- A. Delta Lake
- B. Photon Engine
- C. Unity Catalog ✓**
- D. MLflow

Answer: C | **Why:** Unity Catalog is the governance layer providing a 3-level namespace (catalog.schema.table) and centralized access control, lineage, and auditing across workspaces.

Q3 FOUNDATIONAL

What is the difference between an All-Purpose cluster and a Job cluster?

Domain 1: Databricks Intelligence Platform

- A. All-purpose clusters are cheaper
- B. Job clusters are persistent and shared across users
- C. All-purpose clusters are interactive and persistent; Job clusters auto-terminate after job completion ✓**
- D. Job clusters support more node types

Answer: C | **Why:** All-purpose clusters remain running for interactive notebooks. Job clusters are spun up for a specific job run and terminated afterward, making them more cost-efficient for production.

Q4 FOUNDATIONAL

Where does Delta Lake store its transaction log?

Domain 1: Databricks Intelligence Platform

- A. In a separate Hive Metastore database
- B. In a _delta_log folder at the root of the table directory ✓**
- C. In DBFS root
- D. In the Unity Catalog metadata store

Answer: B | **Why:** Delta Lake writes all transactions as JSON files in the _delta_log directory within the table storage location. This is the source of truth for the table's state.

Q5 FOUNDATIONAL

Which Delta Lake feature allows you to query a table as it existed 5 versions ago?

Domain 1: Databricks Intelligence Platform

- A. Schema Evolution
- B. VACUUM
- C. Time Travel ✓**
- D. Z-Ordering

Answer: C | **Why:** Delta Lake Time Travel allows querying historical data using VERSION AS OF or TIMESTAMP AS OF clauses, powered by the immutable transaction log.

Q6 FOUNDATIONAL

A user runs VACUUM employees RETAIN 0 HOURS. What happens?

Domain 1: Databricks Intelligence Platform

- A. All files are permanently deleted
- B. It deletes all files not referenced by the current table version
- C. It fails unless retentionDurationCheck is disabled
- D. Both B and C ✓

Answer: D | **Why:** VACUUM removes unreferenced files older than the retention period. RETAIN 0 HOURS bypasses the 7-day safety check and requires disabling delta.retentionDurationCheck.enabled config.

Q7 FOUNDATIONAL

What does OPTIMIZE with ZORDER do?

Domain 1: Databricks Intelligence Platform

- A. Deletes duplicate rows
- B. Compacts small files and co-locates related data for faster filtering ✓
- C. Adds indexes to Delta tables
- D. Partitions data by the specified column

Answer: B | **Why:** OPTIMIZE compacts many small Parquet files into fewer large files. ZORDER arranges data so rows with same ZORDER column values are physically close — improving filter performance via data skipping.

Q8 FOUNDATIONAL

Which of the following is TRUE about Delta Lake schema enforcement?

Domain 1: Databricks Intelligence Platform

- A. It allows writing data with extra columns by default
- B. It rejects writes that don't match the table schema by default ✓
- C. It only applies to streaming writes
- D. It must be manually enabled with a table property

Answer: B | **Why:** Delta Lake enforces schema by default. Any write with extra columns or wrong types will be rejected unless mergeSchema or overwriteSchema is explicitly set.

Q9 FOUNDATIONAL

A data engineer wants to add new columns to an existing Delta table during a write. What option should they use?

Domain 2: Data Ingestion & Loading

- A. `.option('overwriteSchema','true')`
- B. `.option('mergeSchema','true')` ✓
- C. `.option('schema.enforce','false')`
- D. ALTER TABLE ADD COLUMNS before writing

Answer: B | **Why:** mergeSchema=true allows new columns in the DataFrame to be automatically added to the Delta table schema during the write, without replacing existing columns or data.

Q10 FOUNDATIONAL

What is the Photon engine in Databricks?

Domain 1: Databricks Intelligence Platform

- A. A streaming engine for Kafka
- B. A C++-based vectorized query engine that speeds up SQL and DataFrame workloads ✓
- C. A GPU-accelerated ML training framework
- D. An alternative to Delta Lake

Answer: B | **Why:** Photon is Databricks native vectorized query engine written in C++. It speeds up SQL and DataFrame operations especially for aggregations and joins, without any code changes.

Q11 FOUNDATIONAL

Which output mode should be used when writing aggregated streaming data to a Delta table?

Domain 4: Working with Lakeflow Jobs

- A. Update mode ✓
- B. Complete mode
- C. Append mode
- D. Overwrite mode

Answer: A | **Why:** Update mode writes only rows that changed in the latest micro-batch. For aggregations with watermarks, update mode is appropriate. Complete mode rewrites the full result set each time.

Q12 FOUNDATIONAL

What is the role of a checkpoint in Structured Streaming?

Domain 4: Working with Lakeflow Jobs

- A. It stores the schema of the streaming DataFrame
- B. It tracks the progress of the stream for fault tolerance and recovery ✓
- C. It schedules micro-batch intervals
- D. It validates data quality

Answer: B | **Why:** Checkpoints store stream progress (offsets, state) to persistent storage. If the stream fails, it resumes from where it left off — critical for exactly-once processing.

Q13 FOUNDATIONAL

A Delta table has 500 small Parquet files. Which command improves query performance?

Domain 1: Databricks Intelligence Platform

- A. VACUUM
- B. ANALYZE TABLE
- C. OPTIMIZE ✓
- D. REFRESH TABLE

Answer: C | **Why:** OPTIMIZE compacts small files into fewer larger files (up to 1GB). This reduces file listing overhead and enables more efficient Parquet column pruning during reads.

Q14 FOUNDATIONAL

What does DESCRIBE HISTORY do on a Delta table?

Domain 1: Databricks Intelligence Platform

- A. Shows the table schema
- B. Lists all columns with data types
- C. Returns a log of all operations (writes, updates, deletes) performed on the table ✓
- D. Shows partition information

Answer: C | **Why:** DESCRIBE HISTORY returns the transaction log as a readable table: version number, timestamp, operation type, user, cluster ID, and other details for every transaction.

Q15 FOUNDATIONAL

Which file format does Delta Lake use for its underlying data storage?

Domain 1: Databricks Intelligence Platform

- A. ORC
- B. Avro
- C. Parquet ✓
- D. CSV

Answer: C | **Why:** Delta Lake stores data as Parquet files. The Delta layer adds a transaction log (_delta_log) on top of Parquet to provide ACID guarantees, schema enforcement, and time travel.

Q16 FOUNDATIONAL

What is the 3-level namespace in Unity Catalog?

Domain 7: Governance & Security

- A. Database.Schema.Table
- B. Catalog.Schema.Table ✓
- C. Workspace.Database.Table
- D. Environment.Catalog.Schema

Answer: B | **Why:** Unity Catalog uses catalog.schema.table, e.g. main.silver.customers. This replaces the legacy two-level hive_metastore.schema.table format.

Q17 FOUNDATIONAL

Which command shows a user's privileges on a Delta table in Unity Catalog?

Domain 7: Governance & Security

- A. DESCRIBE TABLE
- B. LIST GRANTS
- C. SHOW GRANTS ON TABLE catalog.schema.table ✓
- D. SHOW PERMISSIONS table

Answer: C | **Why:** SHOW GRANTS ON TABLE displays all privileges granted on that object, including the grantee and privilege type.

Q18 FOUNDATIONAL

What is a Managed Table in Databricks?

Domain 7: Governance & Security

- A. A table stored in an external cloud bucket
- B. **A table where Databricks manages both metadata and data files; dropping the table deletes the data** ✓
- C. A table with row-level security enabled
- D. A read-only table

Answer: B | **Why:** With managed tables, Databricks controls the data location. DROP TABLE permanently deletes both metadata and underlying Parquet data files.

Q19 FOUNDATIONAL

A data engineer needs to run a notebook on a schedule every day at 8 AM. What Databricks feature should be used?

Domain 4: Working with Lakeflow Jobs

- A. Delta Live Tables
- B. Auto Loader
- C. **Databricks Jobs** ✓
- D. SQL Warehouse

Answer: C | **Why:** Databricks Jobs (Lakeflow Jobs) allow scheduling tasks on a cron schedule or event trigger with configurable retries, notifications, and cluster settings.

Q20 FOUNDATIONAL

Which of the following is NOT a benefit of Delta Live Tables?

Domain 3: Data Transformation & Modeling

- A. Automatic data quality enforcement with expectations
- B. Declarative pipeline definition
- C. **Manual orchestration of each transformation step** ✓
- D. Built-in monitoring and auto-scaling

Answer: C | **Why:** DLT is declarative — you define what you want, not how to execute. Databricks handles orchestration, dependency resolution, and scaling automatically.

Q21 FOUNDATIONAL

What does CTAS (CREATE TABLE AS SELECT) do in Databricks SQL?

Domain 3: Data Transformation & Modeling

- A. Creates an empty table with a schema
- B. **Creates a new table and populates it with the results of a SELECT query** ✓
- C. Creates a temp view
- D. Creates a view from an existing table

Answer: B | **Why:** CTAS creates and populates a table in one step. Schema is inferred from the SELECT result. Use CREATE OR REPLACE TABLE to overwrite if the table already exists.

Q22 FOUNDATIONAL

What is the difference between INSERT INTO and INSERT OVERWRITE?

Domain 3: Data Transformation & Modeling

- A. INSERT INTO is only for external tables
- B. **INSERT OVERWRITE removes existing data and replaces with new data; INSERT INTO appends** ✓
- C. INSERT INTO supports partitioning; INSERT OVERWRITE does not
- D. They are identical in behavior

Answer: B | **Why:** INSERT INTO appends new rows. INSERT OVERWRITE replaces the entire table (or partition) with the new data, effectively truncating and reloading.

Q23 FOUNDATIONAL

Which function handles NULL values by replacing them with a fallback value?

Domain 3: Data Transformation & Modeling

- A. NULLIF
- B. NVL2
- C. **COALESCE** ✓
- D. IFNULL only

Answer: C | **Why:** COALESCE returns the first non-null value in a list of arguments. More versatile than IFNULL (which only takes 2 arguments).

Q24 FOUNDATIONAL

A SQL UDF is created in schema main.analytics. Which user can execute it by default?

Domain 7: Governance & Security

- A. Any user in the workspace
- B. Only the owner
- C. Any user granted EXECUTE privilege on the function ✓
- D. Any user with SELECT on any table in the schema

Answer: C | **Why:** SQL UDFs in Unity Catalog are securable objects. Users must be explicitly granted EXECUTE privilege to call the function.

Q25 FOUNDATIONAL

Which PySpark method registers a DataFrame as a temporary view usable in Spark SQL?

Domain 3: Data Transformation & Modeling

- A. df.registerTable()
- B. df.createOrReplaceTempView('name') ✓
- C. df.toSQL('name')
- D. spark.registerView(df,'name')

Answer: B | **Why:** createOrReplaceTempView() registers the DataFrame as a temp SQL view scoped to the current SparkSession. Query it with spark.sql('SELECT * FROM name').

Q26 FOUNDATIONAL

What does NULLIF(expr1, expr2) return?

Domain 3: Data Transformation & Modeling

- A. The first non-null value
- B. NULL if expr1 equals expr2, otherwise returns expr1 ✓
- C. expr2 if expr1 is null
- D. Always returns NULL

Answer: B | **Why:** NULLIF returns NULL when both expressions are equal. Useful for converting sentinel values (empty strings, zeros) to NULL for proper NULL handling downstream.

Q27 FOUNDATIONAL

Which write option allows adding new columns from a DataFrame to an existing Delta table schema?

Domain 2: Data Ingestion & Loading

- A. overwriteSchema
- B. appendSchema
- C. mergeSchema ✓
- D. evolveSchema

Answer: C | **Why:** The mergeSchema option (.option('mergeSchema','true')) adds new columns from the source DataFrame to the existing Delta table schema without replacing existing data.

Q28 FOUNDATIONAL

What does a Global Temp View differ from a regular Temp View?

Domain 3: Data Transformation & Modeling

- A. It is stored in Delta format
- B. It persists across all sessions in the same cluster; regular Temp Views are session-scoped ✓
- C. It is visible to all workspaces
- D. It supports schema enforcement

Answer: B | **Why:** Global Temp Views live in the global_temp database and are accessible by all sessions on the same cluster. Regular Temp Views are only available in the current SparkSession.

Q29 FOUNDATIONAL

How do you drop a column from a Delta table?

Domain 3: Data Transformation & Modeling

- A. DELETE COLUMN col FROM table
- B. ALTER TABLE table DROP COLUMN col ✓
- C. REMOVE COLUMN col FROM table
- D. Delta tables don't support dropping columns

Answer: B | **Why:** ALTER TABLE table DROP COLUMN column_name is correct. Note: column mapping must be enabled (delta.columnMapping.mode = 'name') for this to work.

Q30 FOUNDATIONAL

A data engineer writes `df.write.mode('overwrite').option('overwriteSchema', true).saveAsTable('sales')`. What happens?

Domain 2: Data Ingestion & Loading

- A. Appends new data and updates schema
- B. Replaces table data and updates the schema to match the DataFrame ✓
- C. Only updates schema without writing data
- D. Fails because overwriteSchema requires a MERGE

Answer: B | **Why:** overwrite mode replaces table data. overwriteSchema=true also replaces the schema with the DataFrame schema. Both data and schema are overwritten.

Q31 FOUNDATIONAL

What is the output of: `SELECT COALESCE(NULL, NULL, 'Karthick', 'Y')`?

Domain 3: Data Transformation & Modeling

- A. NULL
- B. Y
- C. Karthick ✓
- D. Error

Answer: C | **Why:** COALESCE returns the first non-null value. First two args are NULL, so it returns 'Karthick'. 'Y' is ignored because a non-null value was already found.

Q32 FOUNDATIONAL

Which SQL command reshapes rows into columns?

Domain 3: Data Transformation & Modeling

- A. UNPIVOT
- B. RESHAPE
- C. PIVOT ✓
- D. TRANSPOSE

Answer: C | **Why:** PIVOT rotates row values into columns — useful for cross-tab reports. UNPIVOT does the reverse, converting columns back to rows.

Q33 FOUNDATIONAL

What does the following return: `SELECT NULLIF(100, 100)`?

Domain 3: Data Transformation & Modeling

- A. 100
- B. 0
- C. NULL ✓
- D. Error

Answer: C | **Why:** NULLIF returns NULL when both arguments are equal. Since $100 = 100$, the result is NULL.

Q34 FOUNDATIONAL

A data engineer wants to combine two DataFrames vertically (stack rows). Which method is correct?

Domain 3: Data Transformation & Modeling

- A. `df1.join(df2)`
- B. `df1.union(df2)` ✓
- C. `df1.merge(df2)`
- D. `df1.append(df2)`

Answer: B | **Why:** union() combines two DataFrames with the same schema by stacking all rows including duplicates (UNION ALL). Use unionByName() if columns may be in different order.

Q35 FOUNDATIONAL

Which of the following correctly creates a SQL UDF in Databricks?

Domain 3: Data Transformation & Modeling

- A. `CREATE FUNCTION f(x INT) AS x * 2`
- B. `CREATE OR REPLACE FUNCTION double_val(x INT) RETURNS INT RETURN x * 2` ✓
- C. `def double_val(x): return x * 2`
- D. `CREATE UDF double_val(INT) RETURN INT AS x*2`

Answer: B | **Why:** Correct SQL UDF syntax: `CREATE OR REPLACE FUNCTION name(param TYPE) RETURNS TYPE RETURN expression`.

Q36 FOUNDATIONAL**What does `spark.sql()` return?**

Domain 3: Data Transformation & Modeling

- A. A string result
- B. A Python list
- C. **A Spark DataFrame ✓**
- D. A Pandas DataFrame

Answer: C | **Why:** `spark.sql()` executes a SQL query and returns the result as a Spark DataFrame, which can be further transformed or written.**Q37** FOUNDATIONAL**Which is the correct way to pass a Python variable into a Spark SQL query?**

Domain 3: Data Transformation & Modeling

- A. `spark.sql('SELECT * FROM $table')`
- B. `spark.sql(f'SELECT * FROM {table_name}')`
- C. `spark.sql('SELECT * FROM ' + table_name)`
- D. **Both B and C ✓**

Answer: D | **Why:** Both f-strings and string concatenation work for embedding Python variables in SQL strings. F-strings are preferred for readability.**Q38** FOUNDATIONAL**What does `DESCRIBE EXTENDED table_name` show?**

Domain 3: Data Transformation & Modeling

- A. Only column names and types
- B. **Column names, types, plus table metadata (location, format, partition info, owner) ✓**
- C. Only partition information
- D. Only row count

Answer: B | **Why:** `DESCRIBE EXTENDED` provides full table details: column schema, partitioning, storage location, table type (managed/external), file format, table properties, and owner.**Q39** FOUNDATIONAL**A data engineer needs to remove duplicate rows from a Delta table. Which approach is best?**

Domain 3: Data Transformation & Modeling

- A. `DELETE FROM table WHERE rownum > 1`
- B. `CREATE OR REPLACE TABLE t AS SELECT DISTINCT * FROM t`
- C. `INSERT OVERWRITE SELECT DISTINCT`
- D. **Both B and C are valid ✓**

Answer: D | **Why:** Both approaches work: CTAS with `DISTINCT` (B) is clean; `INSERT OVERWRITE` with `SELECT DISTINCT` (C) also works for Delta tables.**Q40** FOUNDATIONAL**Which function extracts the year from a date column in Spark SQL?**

Domain 3: Data Transformation & Modeling

- A. `DATE_PART('year', col)`
- B. `YEAR(col)`
- C. `EXTRACT(YEAR FROM col)`
- D. **All of the above ✓**

Answer: D | **Why:** All three are valid in Spark SQL. `YEAR(col)` is most concise. `EXTRACT` and `DATE_PART` are ANSI-standard alternatives.**Q41** FOUNDATIONAL**What is the result of: `df.filter(col('salary') > 50000).groupBy('dept').count()`?**

Domain 3: Data Transformation & Modeling

- A. Counts all employees per department
- B. **Counts employees earning more than 50000 per department ✓**
- C. Raises an error because filter must come after groupBy
- D. Returns the total count only

Answer: B | **Why:** `filter()` is applied first (lazy evaluation), keeping only salary > 50000 rows. Then `groupBy('dept').count()` counts those filtered rows by department.

Q42 FOUNDATIONAL**Which Spark write mode raises an error if the table already exists?***Domain 2: Data Ingestion & Loading*

- A. append
- B. overwrite
- C. **errorIfExists** ✓
- D. ignore

Answer: C | **Why:** errorIfExists (the default Spark write mode) throws an AnalysisException if the target table or file path already exists.**Q43** FOUNDATIONAL**What does withColumn() do in PySpark?***Domain 3: Data Transformation & Modeling*

- A. Renames an existing column
- B. **Adds a new column or replaces an existing column with the same name** ✓
- C. Drops a column
- D. Reorders columns

Answer: B | **Why:** withColumn(name, expr) adds a new column or replaces the existing column with that name. Returns a new immutable DataFrame.**Q44** FOUNDATIONAL**A data engineer needs to apply a Python lambda function to each row. What should they use?***Domain 3: Data Transformation & Modeling*

- A. df.apply()
- B. df.map()
- C. **df.rdd.map() or a Pandas UDF** ✓
- D. df.foreach()

Answer: C | **Why:** Native Python lambdas require converting to RDD (df.rdd.map()) or using Pandas UDFs for better performance. Avoid RDD operations when possible.**Q45** FOUNDATIONAL**Which command refreshes metadata of an external table whose underlying data changed?***Domain 2: Data Ingestion & Loading*

- A. SYNC TABLE
- B. RELOAD TABLE
- C. **REFRESH TABLE** ✓
- D. UPDATE METADATA

Answer: C | **Why:** REFRESH TABLE tells Spark to reload the table metadata and invalidate cached data, ensuring queries reflect the current state of external files.**Q46** FOUNDATIONAL**What is the purpose of a watermark in Structured Streaming?***Domain 4: Working with Lakeflow Jobs*

- A. It controls micro-batch size
- B. It sets the maximum retention time for checkpoints
- C. **It defines a time threshold for handling late-arriving data in aggregations** ✓
- D. It encrypts streaming data in transit

Answer: C | **Why:** Watermarks bound the state kept for aggregations. Data arriving later than the watermark threshold is dropped, preventing unbounded state growth in long-running streams.**Q47** FOUNDATIONAL**Which trigger processes all available data once and then stops the stream?***Domain 4: Working with Lakeflow Jobs*

- A. Trigger.ProcessingTime
- B. **Trigger.Once() or Trigger.AvailableNow()** ✓
- C. Trigger.Continuous
- D. Trigger.Batch

Answer: B | **Why:** Trigger.Once() processes all available data in one micro-batch and stops. Trigger.AvailableNow() is the improved version using multiple micro-batches with better scaling.

Q48 FOUNDATIONAL**What format should be used in readStream to use Auto Loader?**

Domain 2: Data Ingestion & Loading

- A. delta
- B. kafka
- C. cloudFiles ✓
- D. autoLoader

Answer: C | **Why:** Auto Loader is invoked using `.format('cloudFiles')`. Specify the actual file format with `.option('cloudFiles.format', 'json/csv/parquet/etc')`.**Q49** FOUNDATIONAL**What does the schemaLocation option in Auto Loader do?**

Domain 2: Data Ingestion & Loading

- A. Specifies where to load raw data from
- B. Stores the inferred schema so it persists across restarts ✓
- C. Sets where the output Delta table is written
- D. Controls schema enforcement strictness

Answer: B | **Why:** Auto Loader infers schema from the first batch and stores it at schemaLocation. On subsequent runs it reuses this schema, ensuring consistency without re-inference.**Q50** FOUNDATIONAL**In a medallion architecture, what is the Silver layer responsible for?**

Domain 3: Data Transformation & Modeling

- A. Raw, unprocessed data ingestion
- B. Aggregated business KPIs for dashboards
- C. Cleaned, filtered, and enriched data conforming to a standard schema ✓
- D. Only ML feature engineering

Answer: C | **Why:** Silver is the cleaned/conformed layer: filtered, null-handled, typed, and made consistent. Gold is the aggregation layer for dashboards and KPIs.**Q51** FOUNDATIONAL**Which Delta Live Tables decorator enforces a data quality rule that drops violating rows?**

Domain 3: Data Transformation & Modeling

- A. @dlt.expect
- B. @dlt.expect_or_fail
- C. @dlt.expect_or_drop ✓
- D. @dlt.constraint

Answer: C | **Why:** @dlt.expect_or_drop drops rows violating the expectation and continues the pipeline. @dlt.expect logs violations but keeps rows. @dlt.expect_or_fail stops the pipeline.**Q52** FOUNDATIONAL**What happens when a Structured Streaming query loses its checkpoint?**

Domain 4: Working with Lakeflow Jobs

- A. It resumes from where it left off
- B. It starts reprocessing from the beginning of the source ✓
- C. It fails permanently
- D. It skips to the latest offset automatically

Answer: B | **Why:** Without a checkpoint, the stream has no record of progress and restarts from the beginning. Checkpoints are critical for exactly-once processing.**Q53** FOUNDATIONAL**Which output mode rewrites the full aggregation result to the sink on every micro-batch?**

Domain 4: Working with Lakeflow Jobs

- A. Append
- B. Update
- C. Complete ✓
- D. Overwrite

Answer: C | **Why:** Complete mode writes the entire result table (all aggregated rows) on every trigger. Used with aggregations — typically writing to memory or Delta sinks.

Q54 FOUNDATIONAL

What does readStream return in PySpark?

Domain 4: Working with Lakeflow Jobs

- A. A static DataFrame
- B. **A streaming DataFrame ✓**
- C. A Python generator
- D. A Kafka consumer object

Answer: B | **Why:** spark.readStream returns a streaming DataFrame. Operations on it are lazy and only execute when a writeStream is started with trigger and checkpoint.

Q55 FOUNDATIONAL

A data engineer wants to read only new JSON files arriving in S3. What is the recommended approach?

Domain 2: Data Ingestion & Loading

- A. spark.read.json() in a loop
- B. **Structured Streaming with cloudFiles format (Auto Loader) ✓**
- C. Manual file listing with dbutils.fs.ls()
- D. A Databricks Job running every minute

Answer: B | **Why:** Auto Loader (cloudFiles) is the recommended way to incrementally ingest new files from cloud storage. It tracks processed files and handles millions of files efficiently.

Q56 FOUNDATIONAL

Which property enables schema evolution in Auto Loader?

Domain 2: Data Ingestion & Loading

- A. .option('cloudFiles.inferColumnTypes','true')
- B. **.option('cloudFiles.schemaEvolutionMode','addNewColumns') ✓**
- C. .option('mergeSchema','true')
- D. .option('autoMerge','true')

Answer: B | **Why:** cloudFiles.schemaEvolutionMode=addNewColumns automatically adds new source columns to the target Delta schema when they appear in source files.

Q57 FOUNDATIONAL

What is the Bronze layer in medallion architecture primarily used for?

Domain 3: Data Transformation & Modeling

- A. Business aggregations
- B. Data quality enforcement
- C. **Exact copy of raw source data, preserved as-is ✓**
- D. Serving ML models

Answer: C | **Why:** Bronze stores raw data exactly as received from the source — no transformations. Full auditability and ability to reprocess data if business logic changes.

Q58 FOUNDATIONAL

In DLT, what is a Streaming Table vs a Materialized View?

Domain 3: Data Transformation & Modeling

- A. They are identical
- B. **Streaming Tables process data incrementally; Materialized Views recompute fully on each refresh ✓**
- C. Materialized Views process streaming data; Streaming Tables are batch
- D. Both are always recomputed from scratch

Answer: B | **Why:** DLT Streaming Tables process incrementally — only new records. Materialized Views recompute the full result on each pipeline run.

Q59 FOUNDATIONAL

Which trigger type has the lowest latency in Structured Streaming?

Domain 4: Working with Lakeflow Jobs

- A. ProcessingTime('1 second')
- B. Once
- C. AvailableNow
- D. **Continuous('100 milliseconds') ✓**

Answer: D | **Why:** Trigger.Continuous is designed for sub-second (millisecond) latency, running the stream as a nearly continuous query rather than micro-batches.

Q60 FOUNDATIONAL

A streaming aggregation uses a 5-minute window with a 10-minute watermark. Data arrives 8 minutes late. Is it included?

Domain 4: Working with Lakeflow Jobs

- A. Yes, because 8 < 10 minutes ✓**
- B. No, because 8 > 5 minutes
- C. Yes, always included regardless of watermark
- D. No, late data is never included

Answer: A | **Why:** The watermark is the tolerance for late data. Data arriving 8 minutes late is within the 10-minute watermark, so it is included in the aggregation.

Q61 FOUNDATIONAL

What is the advantage of Trigger.AvailableNow() over Trigger.Once()?

Domain 4: Working with Lakeflow Jobs

- A. AvailableNow processes more files per run
- B. AvailableNow uses multiple micro-batches enabling better auto-scaling and fault tolerance ✓**
- C. AvailableNow is cheaper
- D. They are identical

Answer: B | **Why:** AvailableNow() processes all available data at trigger time in multiple micro-batches — better scaling, checkpointing granularity, and resource optimization vs single-batch Once().

Q62 FOUNDATIONAL

Which DLT expectation stops the pipeline when data quality fails?

Domain 3: Data Transformation & Modeling

- A. @dlt.expect
- B. @dlt.expect_or_drop
- C. @dlt.expect_or_fail ✓**
- D. @dlt.constraint_fail

Answer: C | **Why:** @dlt.expect_or_fail raises an exception and stops the pipeline when any row violates the constraint. Use for critical data quality requirements.

Q63 FOUNDATIONAL

Where should checkpoints be stored for production streaming pipelines?

Domain 4: Working with Lakeflow Jobs

- A. In memory for speed
- B. In a local file system
- C. In a reliable cloud object store (S3, ADLS, GCS) ✓**
- D. In the Delta transaction log

Answer: C | **Why:** Checkpoints must survive cluster restarts — store in cloud storage (S3, ADLS, GCS), not local disk or in-memory.

Q64 FOUNDATIONAL

What is the default VACUUM retention period in Delta Lake?

Domain 1: Databricks Intelligence Platform

- A. 24 hours
- B. 30 days
- C. 7 days ✓**
- D. 90 days

Answer: C | **Why:** VACUUM defaults to 7 days (168 hours) retention. Files newer than this are not deleted. This ensures long-running queries can complete before old files are removed.

Q65 FOUNDATIONAL

A DLT pipeline is set to Continuous mode. What does this mean?

Domain 4: Working with Lakeflow Jobs

- A. It runs once and stops
- B. It runs on a fixed schedule
- C. It runs indefinitely, processing new data as it arrives ✓**
- D. It only processes batch data

Answer: C | **Why:** Continuous mode in DLT keeps the pipeline running permanently, processing data as it arrives with low latency — similar to a long-running streaming query.

Q66 FOUNDATIONAL

Which Auto Loader mode uses cloud storage event notifications for scalable file discovery?

Domain 2: Data Ingestion & Loading

- A. Directory listing mode
- B. File notification mode ✓
- C. Event-driven mode
- D. Polling mode

Answer: B | **Why:** File notification mode uses cloud storage event systems (AWS SNS/SQS, Azure Event Grid) to detect new files instantly. Scales to billions of files better than directory listing.

Q67 FOUNDATIONAL

In Multi-hop architecture, which layer typically feeds ML models and BI dashboards?

Domain 3: Data Transformation & Modeling

- A. Bronze
- B. Silver
- C. Gold ✓
- D. Raw

Answer: C | **Why:** The Gold layer contains aggregated, business-level data — KPIs, summary tables, and feature stores — that feeds dashboards, reports, and ML model training.

Q68 FOUNDATIONAL

What happens if a streaming query is started without a checkpointLocation?

Domain 4: Working with Lakeflow Jobs

- A. It runs fine but cannot recover from failure ✓
- B. Databricks throws an error requiring checkpointLocation
- C. It creates a checkpoint in /tmp automatically
- D. It uses the Delta transaction log as a checkpoint

Answer: A | **Why:** Without checkpointLocation, the stream runs but cannot recover its progress after a failure. Databricks may warn but does not block execution.

Q69 FOUNDATIONAL

What is a SQL Warehouse in Databricks?

Domain 1: Databricks Intelligence Platform

- A. A storage location for SQL files
- B. A compute cluster optimized for running SQL queries, used by Databricks SQL ✓
- C. A data warehouse product separate from Databricks
- D. A collection of SQL notebooks

Answer: B | **Why:** SQL Warehouses are compute resources specifically designed for SQL analytics queries. Separate from all-purpose clusters, optimized for BI tools and SQL workloads.

Q70 FOUNDATIONAL

What task types can be used in a Databricks Job?

Domain 4: Working with Lakeflow Jobs

- A. Only Notebooks
- B. Notebooks, Python scripts, JARs, SQL queries, DLT pipelines ✓
- C. Only DLT pipelines and notebooks
- D. Only Spark Submit jobs

Answer: B | **Why:** Databricks Jobs support: Notebook, Python script, Python wheel, JAR, Spark Submit, SQL, and Delta Live Tables pipeline — providing flexibility for diverse pipeline components.

Q71 FOUNDATIONAL

How do you pass parameters to a Databricks Job notebook task?

Domain 4: Working with Lakeflow Jobs

- A. Using environment variables only
- B. Using dbutils.widgets in the notebook and passing key-value pairs in the job configuration ✓
- C. Using sys.argv
- D. Using a config JSON file

Answer: B | **Why:** Jobs pass parameters as key-value pairs. In the notebook: dbutils.widgets.text('param','default') to define; dbutils.widgets.get('param') to read the value at runtime.

Q72 FOUNDATIONAL

A Databricks Job has Task A and Task B. Task B depends on Task A. What happens if Task A fails?

Domain 4: Working with Lakeflow Jobs

- A. Task B runs regardless
- B. Task B is skipped and marked as skipped
- C. Task B is also marked as failed and does not run ✓**
- D. The entire job restarts from the beginning

Answer: C | **Why:** When a task fails, downstream dependent tasks are not executed and marked as failed/blocked. Use Repair Run to retry only failed tasks after fixing the issue.

Q73 FOUNDATIONAL

What is the purpose of a Databricks SQL Alert?

Domain 4: Working with Lakeflow Jobs

- A. To notify users when a query takes too long
- B. To trigger notifications (email, Slack, webhooks) when a query result meets a specified condition ✓**
- C. To auto-refresh a dashboard
- D. To monitor cluster health

Answer: B | **Why:** Databricks SQL Alerts evaluate a query result against a condition (e.g. count > 1000) on a schedule. If the condition is met, it sends a notification to configured destinations.

Q74 FOUNDATIONAL

What privilege is required to query a table in Unity Catalog?

Domain 7: Governance & Security

- A. USE CATALOG only
- B. SELECT on the table, plus USE SCHEMA and USE CATALOG on parent objects ✓**
- C. MODIFY
- D. READ FILES

Answer: B | **Why:** To query a table: USE CATALOG on the catalog, USE SCHEMA on the schema, and SELECT on the table. All three are required in Unity Catalog.

Q75 FOUNDATIONAL

Which command grants SELECT on a table to a service principal?

Domain 7: Governance & Security

- A. GRANT READ ON TABLE t TO sp
- B. GRANT SELECT ON TABLE catalog.schema.t TO `service-principal-id` ✓**
- C. ALLOW SELECT ON t FOR sp
- D. SET PERMISSION SELECT ON t sp

Answer: B | **Why:** GRANT privilege ON object_type object_name TO principal. Service principals are referenced by their application ID in backticks.

Q76 FOUNDATIONAL

What is the difference between a Serverless SQL Warehouse and a Classic SQL Warehouse?

Domain 1: Databricks Intelligence Platform

- A. Serverless uses Delta Lake; Classic does not
- B. Serverless starts faster and scales instantly without managing compute; Classic requires cluster configuration ✓**
- C. Classic is cheaper
- D. They are identical

Answer: B | **Why:** Serverless SQL Warehouses are fully managed by Databricks with near-instant startup and automatic scaling. Classic warehouses require configuring cluster size and have longer startup times.

Q77 FOUNDATIONAL

A data engineer wants to schedule a SQL dashboard to refresh every hour. Where is this configured?

Domain 4: Working with Lakeflow Jobs

- A. In the cluster configuration
- B. In the dashboard settings under 'Schedule' ✓**
- C. In the SQL Warehouse settings
- D. In a Databricks Job

Answer: B | **Why:** Databricks SQL dashboards have a built-in Schedule option in their settings panel. Configure the refresh frequency and the SQL Warehouse to use.

Q78 FOUNDATIONAL**What does REVOKE MODIFY ON TABLE do?**

Domain 7: Governance & Security

- A. Removes all privileges from the user
- B. Removes the ability to insert, update, and delete rows from the table for that user ✓
- C. Drops the table
- D. Prevents the user from reading the table

Answer: B | **Why:** MODIFY covers INSERT, UPDATE, and DELETE. Revoking it prevents data changes while other privileges (like SELECT) remain unaffected.**Q79** FOUNDATIONAL**In a Databricks Job, what is a Task Cluster vs a Shared Job Cluster?**

Domain 4: Working with Lakeflow Jobs

- A. They are the same
- B. A Task Cluster is created per-task; a Shared Job Cluster is shared across multiple tasks in the same job run ✓
- C. Task Clusters are all-purpose; Shared Job Clusters are serverless
- D. Shared Job Clusters are cheaper per task

Answer: B | **Why:** You can define a job cluster shared by multiple tasks in the same job run to reduce startup overhead, or give each task its own dedicated cluster for isolation.**Q80** FOUNDATIONAL**What is the purpose of dbutils.secrets in Databricks?**

Domain 4: Working with Lakeflow Jobs

- A. To encrypt Delta tables
- B. To securely retrieve secrets (API keys, passwords) from a secret scope without hardcoding them in notebooks ✓
- C. To manage cluster credentials
- D. To store widget values

Answer: B | **Why:** dbutils.secrets.get(scope, key) retrieves secrets stored in Databricks Secret Scopes at runtime. Secrets are never displayed in notebook output.**Q81** FOUNDATIONAL**Which privilege allows a user to CREATE tables and views within a schema?**

Domain 7: Governance & Security

- A. USE SCHEMA alone
- B. CREATE TABLE alone
- C. Both USE SCHEMA and CREATE TABLE are needed ✓
- D. MODIFY on the schema

Answer: C | **Why:** To create objects in a schema, a user needs USE SCHEMA (to access the schema) AND CREATE TABLE/CREATE VIEW privilege on the schema. Neither alone is sufficient.**Q82** FOUNDATIONAL**What happens when you DROP a Managed Delta table?**

Domain 7: Governance & Security

- A. Only metadata is deleted; data files remain
- B. Both metadata and underlying data files are permanently deleted ✓
- C. The table is moved to a trash folder
- D. It creates a backup before deleting

Answer: B | **Why:** Managed tables give Databricks control over data files. DROP TABLE removes both the UC metadata AND permanently deletes the Parquet files from storage.**Q83** FOUNDATIONAL**A Databricks Job is configured with max_retries = 2. The task fails on the first attempt. How many total attempts are made?**

Domain 4: Working with Lakeflow Jobs

- A. 1
- B. 2
- C. 3 ✓
- D. 4

Answer: C | **Why:** max_retries=2 means 2 retries after the initial attempt — 3 total attempts (1 original + 2 retries).

Q84 FOUNDATIONAL**Which Databricks SQL feature allows you to explore data and create visualizations interactively?**

Domain 1: Databricks Intelligence Platform

- A. Databricks Notebooks
- B. Databricks SQL Editor and Dashboards ✓
- C. MLflow UI
- D. Delta Live Tables UI

Answer: B | **Why:** Databricks SQL provides a SQL editor for writing/running queries and a dashboard builder for visualizations — optimized for analysts without notebook overhead.

Q85 FOUNDATIONAL**What does the GRANT USAGE privilege allow in Unity Catalog?**

Domain 7: Governance & Security

- A. Allows running SQL queries on a warehouse
- B. Same as USE SCHEMA or USE CATALOG — allows navigating/accessing the object ✓
- C. Allows creating new objects in a schema
- D. Allows viewing table metadata only

Answer: B | **Why:** USAGE is an alias for USE CATALOG/USE SCHEMA in some versions of Unity Catalog. It grants the ability to access and navigate an object's contents.

Q86 FOUNDATIONAL**What is a Data Object Owner in Unity Catalog?**

Domain 7: Governance & Security

- A. The user who first queries the table
- B. The user or group that created the object; they have all privileges and can grant privileges to others ✓
- C. The admin of the Databricks workspace
- D. The Unity Catalog service account

Answer: B | **Why:** The owner of a data object has full privileges and can grant/revoke privileges to other users or groups. Ownership can be transferred with ALTER OWNER TO.

Q87 FOUNDATIONAL**Which job trigger type runs a job at a fixed time using a cron expression?**

Domain 4: Working with Lakeflow Jobs

- A. Manual trigger
- B. File arrival trigger
- C. Scheduled trigger ✓
- D. Continuous trigger

Answer: C | **Why:** Scheduled triggers use cron syntax to run jobs at fixed times. Databricks also supports file arrival (new file in path) and table update (new data in table) triggers.

Q88 FOUNDATIONAL**A SQL Alert is set to trigger when COUNT > 500. The query returns 450 rows. What happens?**

Domain 4: Working with Lakeflow Jobs

- A. Alert fires and sends notification
- B. Alert does not fire; no notification sent ✓
- C. Alert fires but with a warning
- D. Alert is paused

Answer: B | **Why:** The alert condition COUNT > 500 is not met (450 < 500). The alert does not trigger and no notification is sent.

Q89 FOUNDATIONAL**What is Row-Level Security in Unity Catalog?**

Domain 7: Governance & Security

- A. Encrypting individual rows
- B. Controlling which rows a user can see using dynamic views with current_user() or IS_MEMBER() checks ✓
- C. A built-in feature enabled with ALTER TABLE
- D. Only available for external tables

Answer: B | **Why:** Row-level security is implemented with dynamic views filtering rows based on the current user's identity using current_user() or group membership functions.

Q90 FOUNDATIONAL

Which is a best practice for Databricks Job cost optimization?

Domain 4: Working with Lakeflow Jobs

- A. Always use all-purpose clusters for all jobs
- B. Use job clusters that auto-terminate after job completion ✓**
- C. Use the largest possible cluster to minimize run time
- D. Disable auto-scaling to prevent cost surprises

Answer: B | **Why:** Job clusters are created specifically for a run and terminate automatically. This prevents idle cluster costs from long-running all-purpose clusters.

Q91 FOUNDATIONAL

A data engineer uses Trigger.ProcessingTime('30 seconds'). How often does the streaming micro-batch run?

Domain 4: Working with Lakeflow Jobs

- A. Every 30 milliseconds
- B. Every 30 seconds ✓**
- C. Every 30 minutes
- D. Only when new data arrives

Answer: B | **Why:** Trigger.ProcessingTime('30 seconds') runs a micro-batch every 30 seconds, regardless of whether new data is available.

Q92 FOUNDATIONAL

What is the function of SHOW GRANTS in Unity Catalog?

Domain 7: Governance & Security

- A. Displays all tables a user can access
- B. Shows all privileges granted on a specific object ✓**
- C. Displays the current user's role
- D. Shows workspace-level admin grants

Answer: B | **Why:** SHOW GRANTS ON [object_type] [object_name] displays all privileges explicitly granted on that object, including the grantee and privilege type.

Q93 FOUNDATIONAL

Which method reads data from a Delta table as a static (batch) DataFrame?

Domain 2: Data Ingestion & Loading

- A. spark.readStream.format('delta').load(path)
- B. spark.read.format('delta').load(path) ✓**
- C. spark.load.delta(path)
- D. delta.read(path)

Answer: B | **Why:** spark.read (without Stream) performs a batch read of the current Delta table snapshot. spark.readStream reads it as a streaming source.

Q94 FOUNDATIONAL

A data engineer wants to ensure a Databricks Job sends an email on failure. Where is this configured?

Domain 4: Working with Lakeflow Jobs

- A. In the cluster configuration
- B. In the notebook code using smtpplib
- C. In the Job settings under Notifications ✓**
- D. In the SQL Warehouse settings

Answer: C | **Why:** Databricks Jobs have a Notifications section in job settings where you configure email addresses to notify on job start, success, or failure.

Q95 FOUNDATIONAL

What is the recommended way to handle secrets in a Databricks production pipeline?

Domain 7: Governance & Security

- A. Hardcode secrets in the notebook
- B. Store in a plain text config file on DBFS
- C. Use Databricks Secret Scopes with dbutils.secrets.get() ✓**
- D. Store in a Delta table with restricted access

Answer: C | **Why:** Databricks Secret Scopes securely store credentials. dbutils.secrets.get('scope','key') retrieves secrets at runtime without exposing them in code or notebook output.

Q96 FOUNDATIONAL

Which dashboard element in Databricks SQL shows a single numeric KPI value?

Domain 1: Databricks Intelligence Platform

- A. Bar chart
- B. Table visualization
- C. Counter widget ✓
- D. Text box

Answer: C | **Why:** The Counter visualization in Databricks SQL Dashboards displays a single numeric value — ideal for KPIs like total sales, active users, or error counts.

Q97 FOUNDATIONAL

What is the effect of granting ALL PRIVILEGES on a schema to a group?

Domain 7: Governance & Security

- A. Only SELECT is granted
- B. The group gets SELECT, MODIFY, CREATE TABLE, CREATE VIEW, and all other schema-level privileges ✓
- C. The group becomes the schema owner
- D. It only applies to future tables, not existing ones

Answer: B | **Why:** ALL PRIVILEGES grants every applicable privilege on the object. For a schema: USE SCHEMA, CREATE TABLE, CREATE VIEW, CREATE FUNCTION, and MODIFY on existing objects.

Q98 FOUNDATIONAL

A SQL query is scheduled every 15 minutes but the SQL Warehouse is stopped. What happens?

Domain 1: Databricks Intelligence Platform

- A. The query is skipped
- B. The warehouse auto-starts, runs the query, then returns to stopped state ✓
- C. The query fails permanently
- D. The schedule is paused

Answer: B | **Why:** SQL Warehouses can be configured to auto-start when a query is triggered. They start, execute the query, then can auto-stop after a configured idle period.

Q99 FOUNDATIONAL

Which is the correct order of privilege inheritance in Unity Catalog?

Domain 7: Governance & Security

- A. Table then Schema then Catalog
- B. Schema then Catalog then Table
- C. Catalog then Schema then Table ✓
- D. They are independent with no inheritance

Answer: C | **Why:** Unity Catalog follows top-down hierarchy: Catalog to Schema to Table. To access a table, a user needs USE CATALOG on the catalog, USE SCHEMA on the schema, and the table-level privilege.

Q101-Q144: 2026 NEW DOMAIN QUESTIONS (CI/CD, Troubleshooting, Advanced Governance)

Q101 2026 NEW

What does enabling delta.enableChangeDataFeed=true on a Delta table allow?

Domain 1: Databricks Intelligence Platform

- A. Time travel queries beyond 7 days
- B. Capturing row-level changes (inserts/updates/deletes) for downstream incremental processing via table_changes() ✓
- C. Automatic schema evolution on writes
- D. Enabling Liquid Clustering

Answer: B | **Why:** Change Data Feed records every row-level operation with a _change_type column (insert/update_preimage/update_postimage/delete). Consumers call table_changes() to read only the delta — no full rescan.

Q102 2026 NEW

Which command enables Liquid Clustering on a new Delta table?

Domain 1: Databricks Intelligence Platform

- A. CREATE TABLE t (...) ZORDER BY (col)
- B. CREATE TABLE t (...) CLUSTER BY (col) ✓
- C. ALTER TABLE t ENABLE LIQUID CLUSTERING BY (col)
- D. CREATE TABLE t (...) PARTITION BY (col) CLUSTER

Answer: B | **Why:** CLUSTER BY (col) in CREATE TABLE enables Liquid Clustering. ZORDER applies via OPTIMIZE...ZORDER BY. Liquid Clustering is more flexible — keys can change without full table rewrite.

Q103 2026 NEW

What is the key advantage of Liquid Clustering over PARTITION BY for a user_id column with 50M distinct values?

Domain 1: Databricks Intelligence Platform

- A. PARTITION BY is faster for lookups
- B. CLUSTER BY handles high-cardinality without creating millions of subdirectories; PARTITION BY creates 50M partitions causing a small-file catastrophe ✓
- C. They perform identically
- D. CLUSTER BY requires less storage

Answer: B | **Why:** PARTITION BY user_id with 50M distinct values = 50M directories, each with tiny files — catastrophic. Liquid Clustering co-locates similar user_id rows in files without directory structure.

Q104 2026 NEW

What is Predictive Optimization in Databricks and which tables does it apply to?

Domain 1: Databricks Intelligence Platform

- A. A query plan cache that predicts execution time
- B. Auto-runs OPTIMIZE and VACUUM on Unity Catalog managed tables based on usage patterns — no manual maintenance needed ✓
- C. A cost prediction tool for cluster sizing
- D. An ML-based query rewriting feature

Answer: B | **Why:** Predictive Optimization monitors table usage and automatically schedules OPTIMIZE+VACUUM. Applies to UC managed Delta tables. Enabled at catalog or schema level.

Q105 2026 NEW

What does the RESTORE TABLE command do in Delta Lake?

Domain 1: Databricks Intelligence Platform

- A. Creates a backup of the current table
- B. Reverts the table to a previous version by writing a new transaction log entry pointing to the historical state ✓
- C. Restores files from DBFS backup
- D. Refreshes the table metadata cache

Answer: B | **Why:** RESTORE TABLE t TO VERSION AS OF n writes a new commit to the Delta log pointing to version n. Previous write history is preserved — time travel can still access all versions.

Q106 2026 NEW

What is the purpose of a cluster pool in Databricks?

Domain 1: Databricks Intelligence Platform

- A. To share data between clusters
- B. To maintain pre-provisioned idle instances that new clusters can claim instantly, reducing startup time ✓
- C. To limit cluster sizes for cost control
- D. To provide GPU acceleration

Answer: B | **Why:** Cluster pools keep warm idle VMs. When a cluster starts, it claims pool instances instantly (seconds) instead of waiting for cloud VM provisioning (2-5 minutes). Reduces job startup latency significantly.

Q107 2026 NEW

What is the difference between the control plane and data plane in Databricks architecture?

Domain 1: Databricks Intelligence Platform

- A. Control plane processes data; data plane stores metadata
- B. Control plane is Databricks-managed (web app, scheduler, metadata); data plane is customer cloud (VMs, storage, Spark execution) — customer data never leaves their cloud** ✓
- C. They are identical environments
- D. Data plane is for streaming; control plane is for batch

Answer: B | **Why:** Control plane = Databricks-managed infrastructure (UI, job scheduler, UC metadata). Data plane = customer's cloud account (worker VMs, cloud storage, actual Spark code execution). Security boundary between them.

Q108 2026 NEW

A data engineer runs COPY INTO on a path with 100 files where 60 were already loaded in a previous run. What happens?

Domain 2: Data Ingestion & Loading

- A. All 100 files are reloaded creating duplicates
- B. Only the 40 new files are loaded — COPY INTO skips the 60 already-tracked files in the Delta log** ✓
- C. The command fails because some files already exist
- D. All 100 files are loaded again

Answer: B | **Why:** COPY INTO tracks loaded files in the Delta transaction log. On re-run it only processes the 40 previously unloaded files. Already-loaded files are skipped — making COPY INTO safely re-runnable.

Q109 2026 NEW

Which ingestion method best handles a source with unpredictable daily volume ranging from 1GB to 100GB?

Domain 2: Data Ingestion & Loading

- A. Fixed cluster COPY INTO on a schedule
- B. Auto Loader with serverless compute — auto-scales based on actual file volume detected** ✓
- C. JDBC batch load with a fixed cluster
- D. Manual notebook execution with manual scaling

Answer: B | **Why:** Auto Loader with serverless compute auto-scales to actual volume. On high-volume days it scales up; on quiet days it scales to minimum. Fixed clusters either over-provision or under-provision for variable volumes.

Q110 2026 NEW

What is cloudFiles.schemaEvolutionMode='rescue' in Auto Loader?

Domain 2: Data Ingestion & Loading

- A. Drops all new columns encountered
- B. Sends schema change alerts to the pipeline owner
- C. Captures new/unexpected columns in the rescuedDataColumn instead of failing or adding to schema** ✓
- D. Auto-creates new schemas for each file version

Answer: C | **Why:** rescue mode captures unexpected/new columns in the rescuedDataColumn JSON field rather than failing (strict) or adding to schema (addNewColumns). A middle-ground option for controlled schema evolution.

Q111 2026 NEW

What is the correct 2026 Lakeflow SDP import for Spark Declarative Pipelines?

Domain 3: Data Transformation & Modeling

- A. import dlt
- B. from pyspark import pipelines as dp** ✓
- C. from databricks import lakeflow
- D. import lakeflow.pipelines

Answer: B | **Why:** The 2026 SDP API uses: from pyspark import pipelines as dp. The old 'import dlt' still works for backwards compatibility but dp is the recommended 2026 namespace.

Q112 2026 NEW

In Lakeflow SDP 2026, which decorator is correct for a KPI table using dp.read() (batch, no streaming)?

Domain 3: Data Transformation & Modeling

- A. @dp.table
- B. @dp.streaming_table
- C. @dp.materialized_view** ✓
- D. @dp.batch_table

Answer: C | **Why:** @dp.materialized_view is for batch-computed tables using dp.read(). @dp.table requires a streaming source (readStream). This decorator distinction is KEY and NEW in the 2026 exam.

Q113 2026 NEW

How are currently active records identified in an SCD Type II table created by `dp.create_auto_cdc_flow()`?

Domain 3: Data Transformation & Modeling

- A. `is_current = True`
- B. `__END_AT IS NULL` ✓
- C. `__START_AT = MAX(__START_AT)`
- D. `scd_version = 'CURRENT'`

Answer: B | **Why:** Lakeflow SDP auto-manages `__START_AT` and `__END_AT` columns. Active records have `__END_AT IS NULL`. When a record changes, the old row gets `__END_AT = change timestamp`, new row inserted with `__END_AT = NULL`.

Q114 2026 NEW

A streaming pipeline fails with `ROW_NUMBER()` not supported on streaming DataFrames. What is the fix?

Domain 3: Data Transformation & Modeling

- A. Use `RANK()` instead
- B. Use `dropDuplicates(['key_column'])` which is streaming-safe ✓
- C. Add `.withWatermark()` before `ROW_NUMBER()`
- D. Use `collect()` to bring data to driver first

Answer: B | **Why:** `ROW_NUMBER()` requires global row visibility — impossible in streaming micro-batches. `dropDuplicates(['key'])` is the streaming-safe replacement. For cross-batch dedup, combine with `withWatermark()`.

Q115 2026 NEW

What is the difference between `repartition(n)` and `coalesce(n)` in PySpark?

Domain 3: Data Transformation & Modeling

- A. They are identical
- B. `repartition` performs a full shuffle and can increase OR decrease partitions; `coalesce` only decreases with minimal data movement (no full shuffle) ✓
- C. `coalesce` is for columns; `repartition` is for rows
- D. `repartition` is streaming-only

Answer: B | **Why:** `repartition(n)`: full shuffle, can increase or decrease — use to fix skew or increase parallelism. `coalesce(n)`: decrease only, no full shuffle — more efficient when only reducing partition count.

Q116 2026 NEW

What is the difference between File Arrival trigger and Table Update trigger in Lakeflow Jobs?

Domain 4: Working with Lakeflow Jobs

- A. They are identical
- B. File Arrival monitors cloud storage for new files; Table Update monitors a Delta table for new committed data — better for data-driven pipeline chaining ✓
- C. File Arrival is for streaming; Table Update for batch
- D. Table Update requires UC; File Arrival does not

Answer: B | **Why:** File Arrival fires when a new file appears at a storage path. Table Update fires when a monitored Delta table receives a new data commit — tighter with the Delta/UC ecosystem for data-driven chaining.

Q117 2026 NEW

What does 'Repair Run' in Lakeflow Jobs do?

Domain 4: Working with Lakeflow Jobs

- A. Deletes the failed job and creates a new identical one
- B. Reruns only failed and skipped tasks while preserving successful task outputs ✓
- C. Restarts all tasks from the beginning
- D. Sends a notification about the failure

Answer: B | **Why:** Repair Run reruns only failed/skipped tasks while keeping outputs from successfully completed tasks. Most efficient recovery method — saves significant compute vs full rerun on long multi-task pipelines.

Q118 2026 NEW

What is a Declarative Automation Bundle (DAB) in Databricks?

Domain 5: Implementing CI/CD

- A. A Docker container for Databricks workloads
- B. **A YAML-based IaC framework for deploying jobs, pipelines, and notebooks across environments (formerly Databricks Asset Bundles)** ✓
- C. A Python package manager for Databricks libraries
- D. A monitoring tool for Databricks clusters

Answer: B | **Why:** DABs (formerly Databricks Asset Bundles) define full resource configurations as YAML files in Git. Enables CI/CD by deploying consistently to dev/test/prod with environment-specific variable overrides.

Q119 2026 NEW

What is the root configuration file for a Declarative Automation Bundle?

Domain 5: Implementing CI/CD

- A. bundle.json
- B. config.yaml
- C. **databricks.yml** ✓
- D. settings.toml

Answer: C | **Why:** databricks.yml is the mandatory root file. It defines bundle name, workspace host, targets (dev/staging/prod), and references to resource YAML definitions (jobs, pipelines, clusters).

Q120 2026 NEW

Which CLI command validates a DAB's YAML without deploying anything?

Domain 5: Implementing CI/CD

- A. databricks bundle check
- B. **databricks bundle validate** ✓
- C. databricks bundle test
- D. databricks bundle lint

Answer: B | **Why:** databricks bundle validate checks all YAML files for syntax errors, invalid configurations, and logical issues — without deploying anything. Always run in CI before deploy.

Q121 2026 NEW

How do DABs handle different cluster sizes in dev vs prod without changing code?

Domain 5: Implementing CI/CD

- A. Maintain separate YAML files per environment
- B. **Use the targets block in databricks.yml with variable overrides — same codebase, different configs per environment** ✓
- C. Use different Git branches per environment
- D. Hardcode environment checks in notebook code

Answer: B | **Why:** DABs targets block: each target (dev/prod) overrides specific variables (catalog name, cluster size). Business logic is identical everywhere — only infrastructure configuration differs via variable overrides.

Q122 2026 NEW

What does 'databricks bundle deploy --target prod' do?

Domain 5: Implementing CI/CD

- A. Runs all prod jobs immediately
- B. Validates YAML syntax only
- C. **Uploads code assets and creates/updates resources (jobs, pipelines) in the prod workspace** ✓
- D. Destroys prod resources and redeploys

Answer: C | **Why:** bundle deploy uploads assets and creates or updates workspace resources (jobs, pipelines) for the specified target. Does NOT run the jobs — use bundle run for execution.

Q123 2026 NEW

What is Databricks Connect?

Domain 5: Implementing CI/CD

- A. A connector between two Databricks workspaces
- B. **An SDK allowing local IDE code (VS Code, PyCharm) to execute on a remote Databricks cluster — enables local debugging with full cluster power** ✓
- C. A feature connecting SQL Warehouses to BI tools
- D. A network peering feature between cloud VPCs

Answer: B | **Why:** Databricks Connect routes local Spark code to a remote Databricks cluster. Developers get IntelliSense, debugger support, and unit testing locally while processing runs on the cluster.

Q124 2026 NEW

A team wants jobs to deploy automatically when code is merged to main branch. Which approach is correct?

Domain 5: Implementing CI/CD

- A. Set a file arrival trigger on the Git repository
- B. Use a Lakeflow Jobs table update trigger
- C. **Integrate DABs with GitHub Actions — runs bundle validate + bundle deploy on merge to main ✓**
- D. Use Databricks Repos auto-sync feature

Answer: C | **Why:** Standard CI/CD: GitHub Actions detects merge to main → runs bundle validate → bundle deploy --target prod. Automates deployment without manual intervention, ensuring only reviewed code reaches production.

Q125 2026 NEW

What does 'databricks bundle destroy' do?

Domain 5: Implementing CI/CD

- A. Deletes all local YAML files
- B. **Removes all Databricks resources (jobs, pipelines) deployed by the bundle from the target workspace, leaving local files intact ✓**
- C. Reverts the workspace to previous bundle version
- D. Deletes the connected Git repository

Answer: B | **Why:** bundle destroy removes only workspace resources created by bundle deploy. Local YAML files and Git repository are unchanged. Use carefully in production — it deletes live jobs and pipelines.

Q126 2026 NEW

What is the recommended DAB folder structure?

Domain 5: Implementing CI/CD

- A. Single databricks.yml with all code
- B. **databricks.yml (root) + resources/ (job/pipeline YAML definitions) + src/ (notebooks, Python scripts) ✓**
- C. A single notebooks/ directory
- D. Separate Git repos per job

Answer: B | **Why:** Best practice: databricks.yml at root, resources/ for infrastructure YAML (job_definition.yml, pipeline_definition.yml), src/ for business logic. Separates infrastructure from code for maintainability.

Q127 2026 NEW

A Spark job shows a few tasks taking 10x longer than others in the Stages tab. What does this indicate?

Domain 6: Troubleshooting, Monitoring & Optimization

- A. Cluster is undersized — add more workers
- B. **Data skew — some partitions have far more data. Fix: enable AQE, add salting to the skewed key, or repartition by a different column ✓**
- C. Network latency — upgrade bandwidth
- D. The Delta table needs VACUUM

Answer: B | **Why:** Task duration imbalance is the classic data skew signature. AQE (spark.sql.adaptive.enabled=true) handles some skew automatically. Manual: salt the skewed join key, filter early, or repartition by a better column.

Q128 2026 NEW

What does high Shuffle Read size in the Spark UI Stages tab indicate?

Domain 6: Troubleshooting, Monitoring & Optimization

- A. Too many input files
- B. **Significant data exchange between executors for join/groupBy — potentially a bottleneck ✓**
- C. Inefficient storage format
- D. Network bandwidth exceeded

Answer: B | **Why:** Large shuffle read means significant data movement across the network for joins, groupBy, and sort. Reduce with: broadcast join for small tables, filter data before shuffling, or adjust shuffle partitions.

Q129 2026 NEW

What is disk spill in Spark and what causes it?

Domain 6: Troubleshooting, Monitoring & Optimization

- A. Writing final output to cloud storage
- B. **When executor memory is insufficient for sort/join/aggregation, Spark writes overflow to local disk — 10-100x slower than in-memory ✓**
- C. Delta checkpoint data written to DBFS
- D. Audit logs overflowing

Answer: B | **Why:** Disk spill occurs when executor heap is exhausted. Spark serialises excess data to local disk. Visible as 'Spill (disk)' in Stages tab. Fix: increase spark.executor.memory, increase numPartitions to reduce per-partition size.

Q130 2026 NEW

What does `spark.sql.adaptive.enabled=true` enable?

Domain 6: Troubleshooting, Monitoring & Optimization

- A. Automatic cluster scaling
- B. Adaptive Query Execution — dynamically adjusts query plans at runtime based on actual data statistics collected during execution ✓**
- C. Automatic schema evolution
- D. Liquid Clustering recluster

Answer: B | **Why:** AQE collects runtime statistics mid-query and dynamically adjusts: coalesces small shuffle partitions (reduces overhead), fixes skew joins, switches between sort-merge and broadcast join automatically.

Q131 2026 NEW

What does the SQL/DataFrame tab in the Spark UI show that is most useful for optimization?

Domain 6: Troubleshooting, Monitoring & Optimization

- A. SQL query edit history
- B. Physical execution plan as a visual DAG with stage boundaries, exchange nodes, join strategies, and row counts at each operation ✓**
- C. Database schema browser
- D. Active SQL connections

Answer: B | **Why:** The SQL/DataFrame tab shows the physical query plan DAG with: file scans (files skipped), exchange/shuffle nodes (bottlenecks), join strategies (broadcast vs sort-merge), and row counts — essential for understanding where time is spent.

Q132 2026 NEW

What is the impact of `spark.sql.shuffle.partitions=200` on a 50MB dataset?

Domain 6: Troubleshooting, Monitoring & Optimization

- A. Optimal — 200 is always the right default
- B. Creates 200 tiny partitions with more scheduling overhead than computation — wasting time and slowing the job ✓**
- C. Causes OOM errors for small data
- D. No impact — Spark ignores this for small datasets

Answer: B | **Why:** 200 shuffle partitions on 50MB = 200 tasks each processing ~250KB. Scheduling overhead (JVM startup, serialisation) exceeds computation. Reduce to 4-8 for small datasets. AQE handles this automatically if enabled.

Q133 2026 NEW

A nightly job that ran in 20 minutes now takes 60 minutes with no code changes. Where should the engineer look first?

Domain 6: Troubleshooting, Monitoring & Optimization

- A. The cluster was secretly resized
- B. Check Lakeflow Jobs run history for duration trend — gradual increase = data growth; sudden spike = skew or infrastructure change. Then check Spark UI for details ✓**
- C. The network connection degraded
- D. The Delta table needs ANALYZE

Answer: B | **Why:** Lakeflow Jobs run history provides duration trends. Gradual increase = data growth. Sudden spike = skew, infrastructure change, or new data pattern. After identifying the start date, check Spark UI for skew/spill.

Q134 2026 NEW

Delta table queries slow after 6 months of micro-batch streaming writes. What maintenance is needed?

Domain 6: Troubleshooting, Monitoring & Optimization

- A. VACUUM only
- B. OPTIMIZE (compact small streaming files, optionally with ZORDER or CLUSTER BY), then VACUUM to remove old files ✓**
- C. REFRESH TABLE
- D. ANALYZE TABLE only

Answer: B | **Why:** Streaming writes create many small files (one per micro-batch). OPTIMIZE compacts them into right-sized files. VACUUM removes the old small files. Always OPTIMIZE before VACUUM — VACUUM removes files OPTIMIZE just replaced.

Q135 2026 NEW

A streaming pipeline batch processing time consistently exceeds the trigger interval. What does this mean?

Domain 6: Troubleshooting, Monitoring & Optimization

- A. The pipeline is running efficiently
- B. The pipeline is falling behind — processing each batch takes longer than the trigger interval, so data accumulates faster than processed ✓**
- C. The checkpoint is corrupted
- D. The watermark is too short

Answer: B | **Why:** When batch processing time > trigger interval, the pipeline falls behind. Solutions: increase cluster resources, optimise transformations (remove unnecessary shuffles), increase trigger interval, or enable Continuous mode.

Q136 2026 NEW

What is the difference between REVOKE and DENY in Unity Catalog?

Domain 7: Governance & Security

- A. They are identical
- B. **REVOKE removes a previously granted privilege; DENY explicitly blocks — overrides GRANT even if user has it via group membership ✓**
- C. REVOKE is for users; DENY is for groups
- D. DENY is permanent; REVOKE is temporary

Answer: B | **Why:** REVOKE removes a GRANT — the user may still have access via group membership. DENY explicitly blocks access regardless of any other grants. If a user has GRANT via group AND a DENY, DENY wins.

Q137 2026 NEW

A company wants to hide salary data from employees while HR managers see full values. Which UC feature implements this?

Domain 7: Governance & Security

- A. Row-level security filter on the salary column
- B. **Column masking — SQL function returning full value for HR managers, NULL for others at query time ✓**
- C. DENY SELECT on the salary column for non-HR users
- D. Encrypt the column at rest

Answer: B | **Why:** Column masking applies a user-defined function at query time: IF(is_member('HR_managers'), salary, NULL). Non-HR users see NULL; HR managers see actual values. NEW in May 2026 exam.

Q138 2026 NEW

What is Unity Catalog ABAC (Attribute-Based Access Control)?

Domain 7: Governance & Security

- A. A Databricks billing model
- B. **Centralised UC policies combining row filters and column masks based on user attributes (group membership, tags) to control fine-grained access across multiple tables ✓**
- C. An access model for cluster-level resources only
- D. A feature for controlling API rate limits

Answer: B | **Why:** ABAC policies define fine-grained access rules based on subject attributes (user groups, service principals) and object attributes (table tags, column sensitivity). Policies apply consistently across all matching tables. NEW May 2026.

Q139 2026 NEW

Only users in the finance_team group should see rows where department=Finance. Which UC feature achieves this?

Domain 7: Governance & Security

- A. Column masking on the department column
- B. **Row-level security (row filter) — boolean function applied at query time filtering rows based on group membership ✓**
- C. Table-level GRANT with WHERE clause
- D. Partition by department and grant per-partition access

Answer: B | **Why:** Row filters apply a filtering function at query time. finance_team members see only Finance rows; others see no Finance rows. Implemented as a UC row filter function or ABAC policy. NEW in May 2026 exam.

Q140 2026 NEW

What is Delta Sharing and what is its key benefit?

Domain 7: Governance & Security

- A. A mechanism to copy Delta tables between Databricks workspaces
- B. **An open protocol sharing live Delta tables externally without copying data — recipients see current data using open-source clients, no Databricks account required ✓**
- C. A backup mechanism for Delta tables
- D. A feature to replicate tables across cloud regions

Answer: B | **Why:** Delta Sharing is zero-copy, live data sharing. Recipients use open-source client libraries (Python/pandas/Spark/Power BI). No data duplication, no stale copies, no Databricks dependency for external recipients.

Q141 2026 NEW

What is Lakehouse Federation in Unity Catalog?

Domain 7: Governance & Security

- A. Federating multiple Databricks workspaces under one governance layer
- B. Creating foreign catalogs in UC that map to external databases (MySQL, PostgreSQL, Snowflake) enabling direct SQL queries without moving data** ✓
- C. Sharing Delta tables across cloud providers
- D. Federating authentication across identity providers

Answer: B | **Why:** Lakehouse Federation creates foreign catalogs pointing to external data sources. Users query with standard SQL as if native UC tables. Data stays in source — no ETL. Predicate pushdown optimises performance. NEW May 2026.

Q142 2026 NEW

What is the cost consideration for Delta Sharing when sharing data across different cloud providers?

Domain 7: Governance & Security

- A. It is free — zero-copy means zero cost
- B. Cross-cloud data access incurs cloud egress fees as data traverses between cloud provider networks — significant for large datasets** ✓
- C. Only the recipient pays egress costs
- D. Delta Sharing requires a cross-cloud license

Answer: B | **Why:** When Azure recipient accesses data shared from AWS, egress fees apply on the AWS provider side. For large datasets this can be substantial. Same-cloud, same-region sharing minimises or eliminates egress costs.

Q143 2026 NEW

How does Unity Catalog data lineage work?

Domain 7: Governance & Security

- A. Engineers manually document lineage in a metadata tool
- B. UC automatically captures column-level lineage across notebooks, SQL queries, jobs, and pipelines — viewable in Catalog Explorer without any code changes** ✓
- C. Lineage requires @lineage decorator annotations in notebook code
- D. Lineage is only table-level, not column-level

Answer: B | **Why:** UC auto-captures lineage as queries run — no instrumentation needed. Records which source columns produce which target columns, across all Databricks surfaces. Column-level lineage in Catalog Explorer.